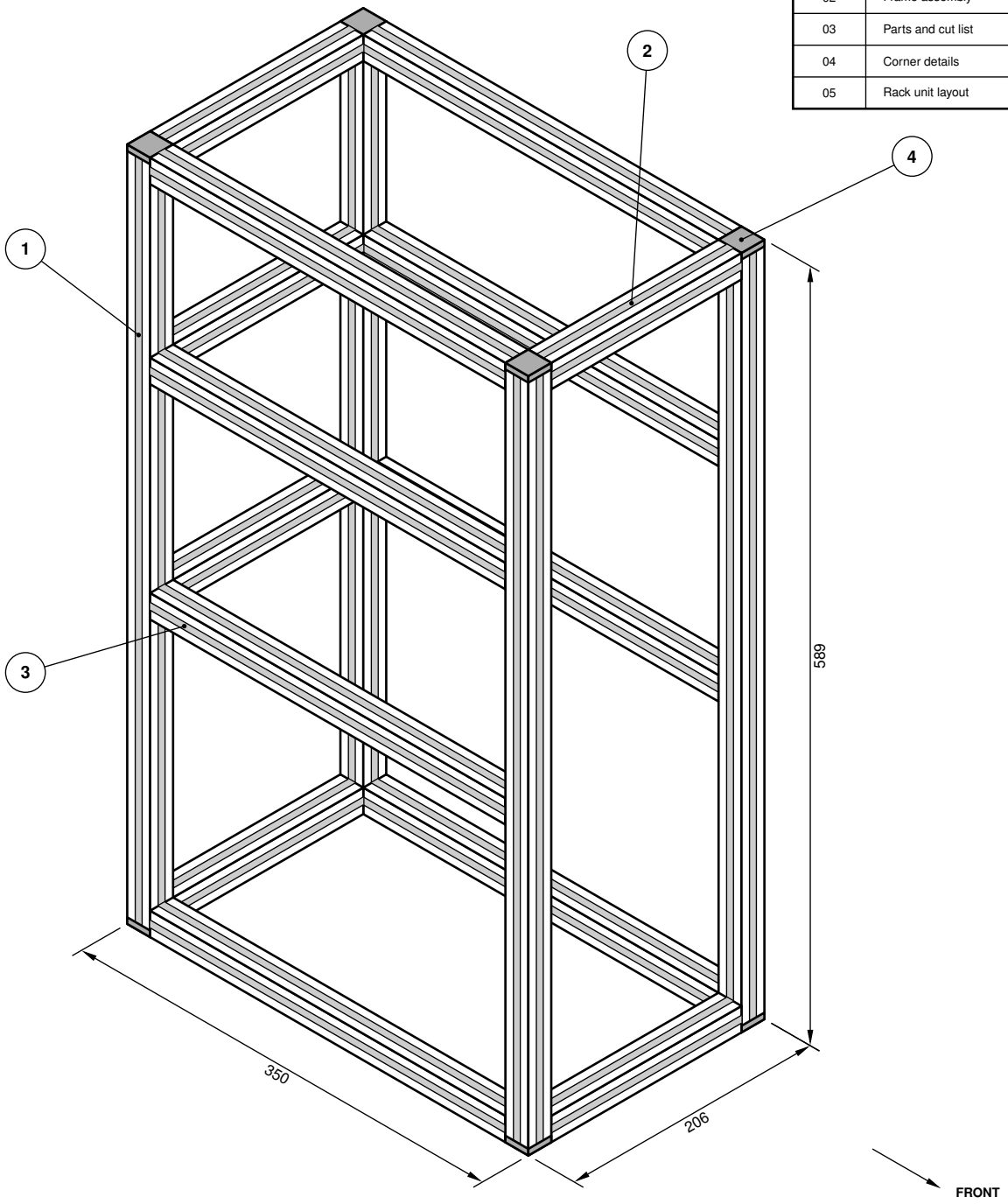
ISOMETRIC VIEW 1:5

SHEET	TITLE
01	Isometric view
02	Frame assembly
03	Parts and cut list
04	Corner details
05	Rack unit layout



NOTES

1. All dimensions are in mm; keep in mind that 1U = 1.75 in = 44.45 mm.
2. Isometric drawing at 1:5 on all three axes. Hidden edges are omitted. Slots shaded in light grey, corner connectors in dark grey.
3. The overall height of 589 includes the 5 mm corner connectors at the top and bottom vertices; uprights are cut to 579.
4. Item numbers match sheets 02 to 04. Orthographic views on sheet 02, cut lengths on sheet 03, corner details on sheet 04, rack unit layout on sheet 05.
5. Intermediate depth rails and back width rails at the U3/U4 and U7/U8 boundaries.

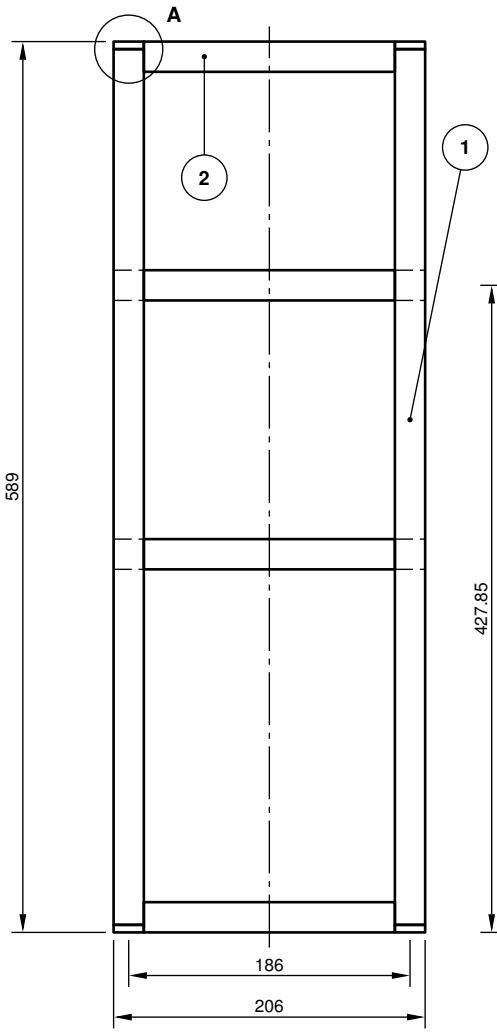
OWNER

LUIDLAB · MINIRACK PROJECT · 8 U. · 20 × 20 ALUMINUM EXTRUSION

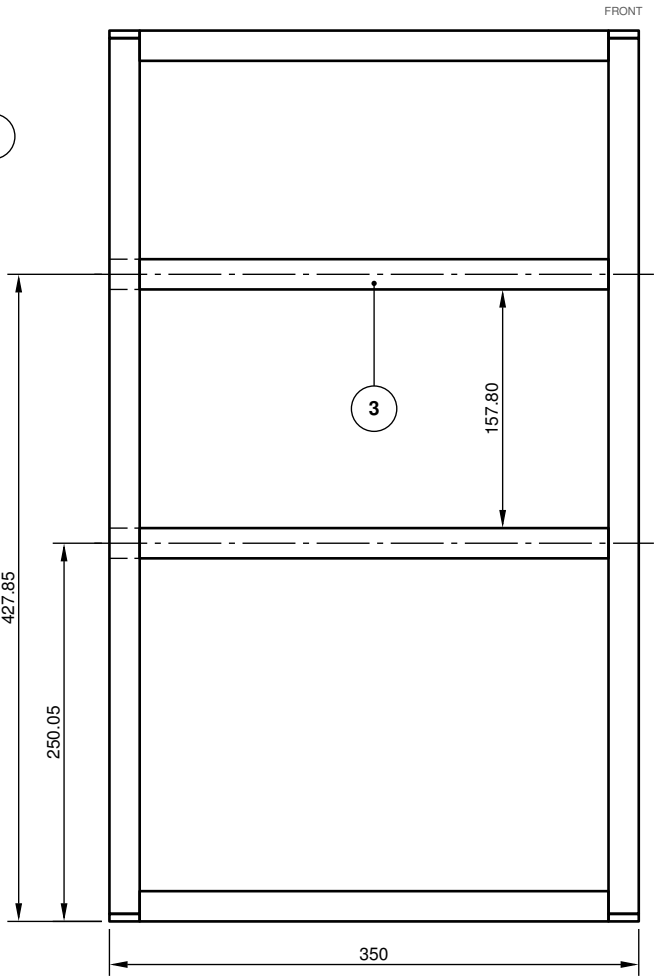
TITLE	PROJECTION	UNIT
ISOMETRIC VIEW Frame overview, 8 in. × 12U format	**ISOMETRIC**	**mm**

MATERIAL	STANDARDS
Aluminum profile 6063-T5, 20 series (20 × 20), 6 mm slot	ISO 5457 · 7200 · 128 · 129-1

DRAWING NO.	SCALE	SHEET	DATE	REV.
MR2020-01	**1:5**	**1 / 5**	2026-09-13	**A**



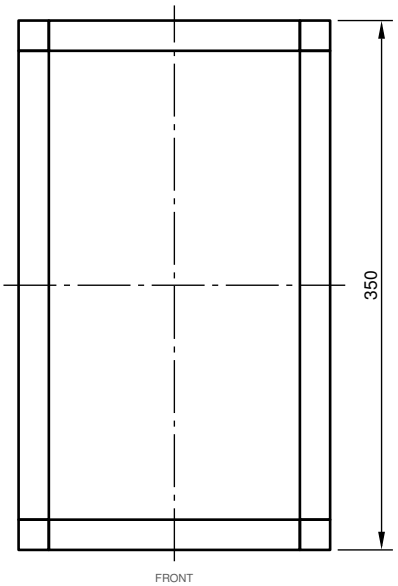
FRONT VIEW



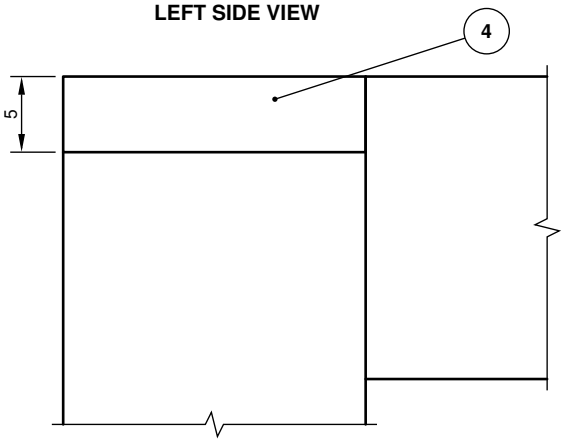
LEFT SIDE VIEW

NOTES

1. All dimensions are in mm; keep in mind that 1U = 1.75 in = 44.45 mm.
2. Heights measured from the frame base, at the bottom face of the lower connectors.
3. Each corner connector adds 5 mm, at the top and at the bottom vertices: uprights are cut to 579 and the frame totals 589. See detail A.
4. First angle projection. The front is on the right in the side view and at the bottom in the top view.
5. Intermediate depth rails and back width rails at the U3/U4 and U7/U8 boundaries.
6. Cut lengths on sheet 03.



TOP VIEW



DETAIL A (2:1)

5 mm connector at each upright end, bottom mirrored

ITEM	DESCRIPTION	QTY	NOTE
1	Upright	4	cut 579
2	Width rail	6	cut 166
3	Depth rail	8	cut 310
4	3-way corner connector	8	purchased

OWNER

LUIDLAB · MINIRACK PROJECT · 8 U. · 20 × 20 ALUMINUM EXTRUSION

TITLE

FRAME ASSEMBLY

20 × 20 extrusion, 8 in. × 12U format

PROJECTION

UNIT

mm

MATERIAL

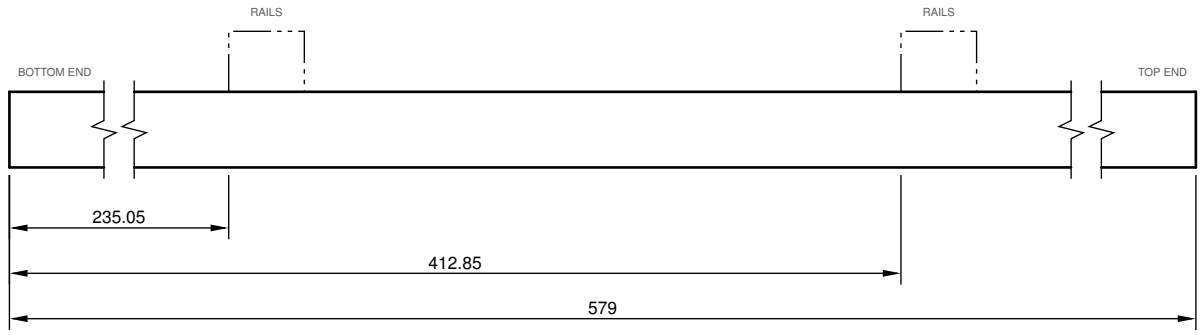
Aluminum profile 6063-T5, 20 series (20 × 20), 6 mm slot

STANDARDS

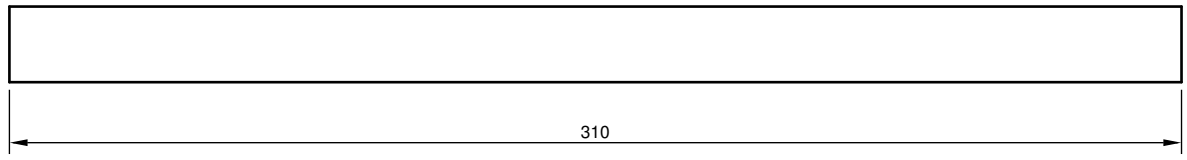
ISO 5457 · 7200 · 128 · 129-1

DRAWING NO.	SCALE	SHEET	DATE	REV.
MR2020-02	**1:5**	**2 / 5**	2026-09-13	**A**

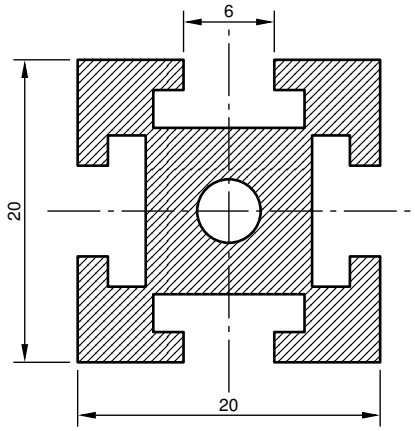
ITEM 1 UPRIGHT QTY 4 — 1:2



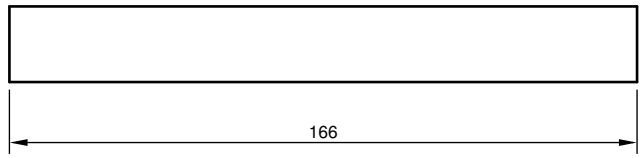
ITEM 3 DEPTH RAIL QTY 8 — 1:2



PROFILE SECTION — 2:1



ITEM 2 WIDTH RAIL QTY 6 — 1:2



CUT LIST

ITEM	DESCRIPTION	QTY	LENGTH	TOTAL
1	Upright	4	579	2316
2	Width rail	6	166	996
3	Depth rail	8	310	2480
			TOTAL PROFILE	**5792**

NOTES

1. All dimensions are in mm. General length tolerance ±0.5.
2. Cut square: the corner connectors depend on it. Deburr both ends of every piece.
3. Lengths do not include saw kerf. Measure before each cut.
4. Upright positions are measured from the bottom end of the upright to the lower face of each rail. Sheet 02 gives the rail axes from the frame base: subtract 5 mm for the lower connector and 10 mm for half a rail.
5. The corner connectors add 5 mm at each end of the upright and are not part of the cut length.
6. On the back uprights the same marks also locate the back intermediate width rails.
7. The upright is drawn as an interrupted view; the dimensions give the true lengths.
8. Profile section is schematic. Slot and bore geometry follow the supplier catalog.

OWNER

LUIDLAB · MINIRACK PROJECT · 8 U. · 20 × 20 ALUMINUM EXTRUSION

TITLE

PARTS AND CUT LIST

Cut pieces for the 8 in. × 12U frame

PROJECTION

UNIT

mm

MATERIAL

Aluminum profile 6063-T5, 20 series (20 × 20), 6 mm slot

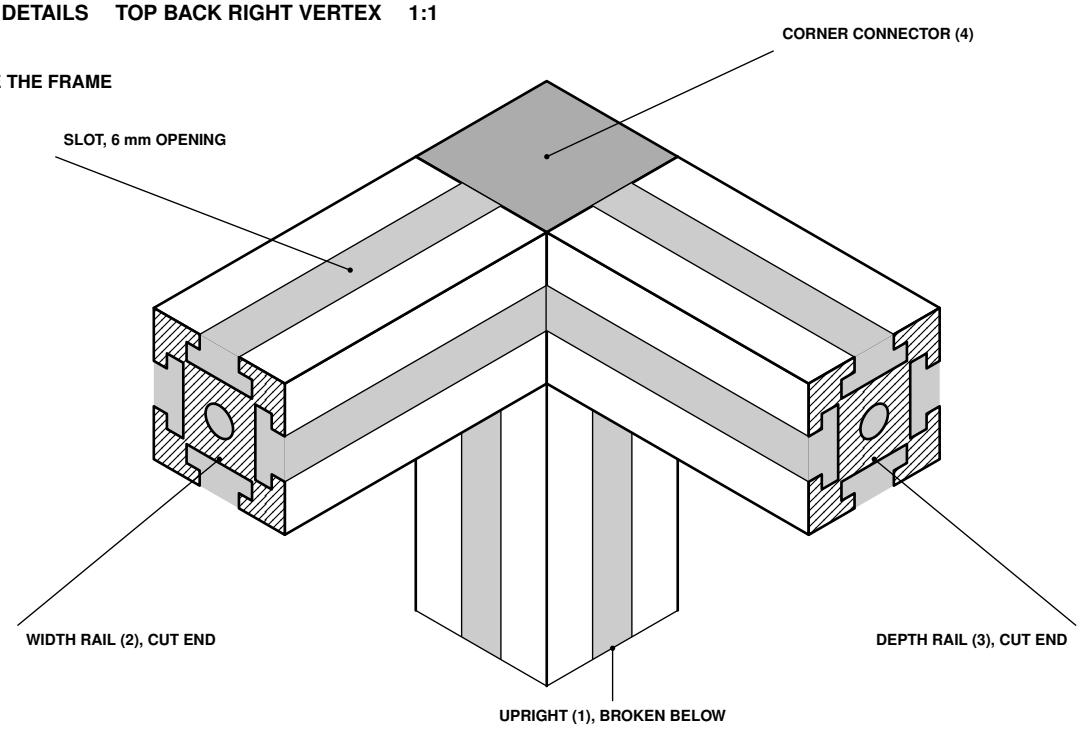
STANDARDS

ISO 5457 · 7200 · 128 · 129-1

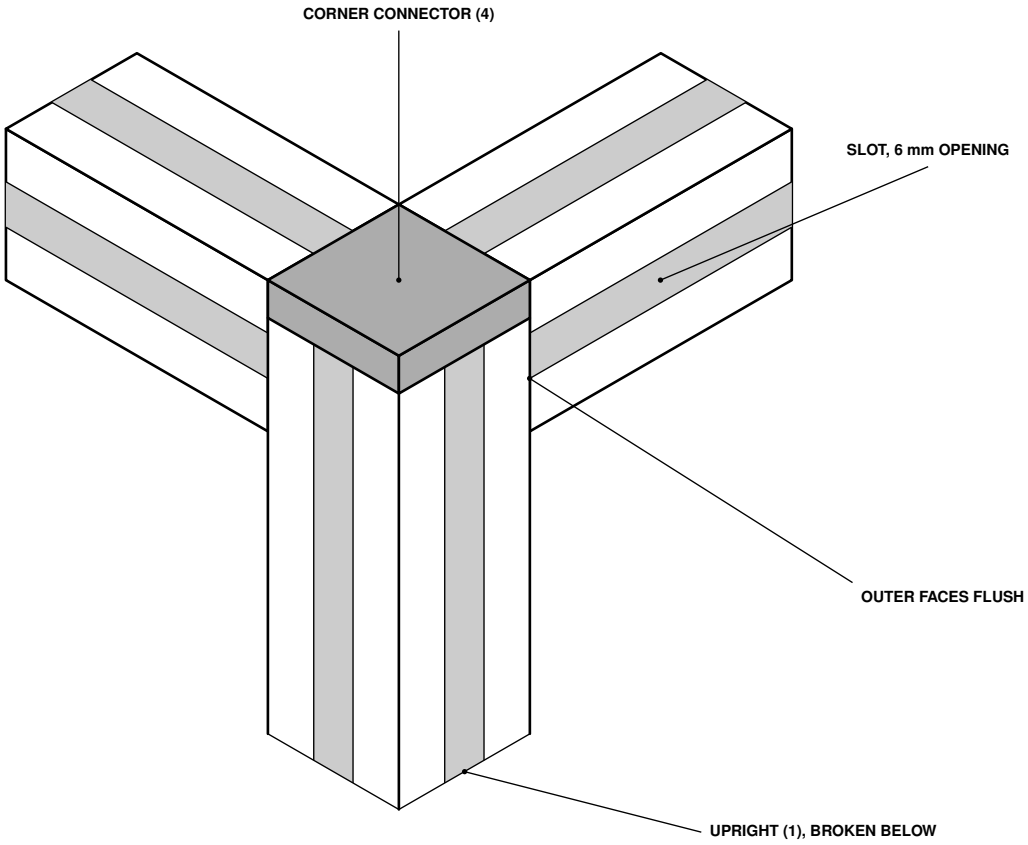
DRAWING NO.	SCALE	SHEET	DATE	REV.
MR2020-03	**1:2 / 2:1**	**3 / 5**	2026-09-13	**A**

CORNER DETAILS TOP BACK RIGHT VERTEX 1:1

A INSIDE THE FRAME



B OUTSIDE THE FRAME



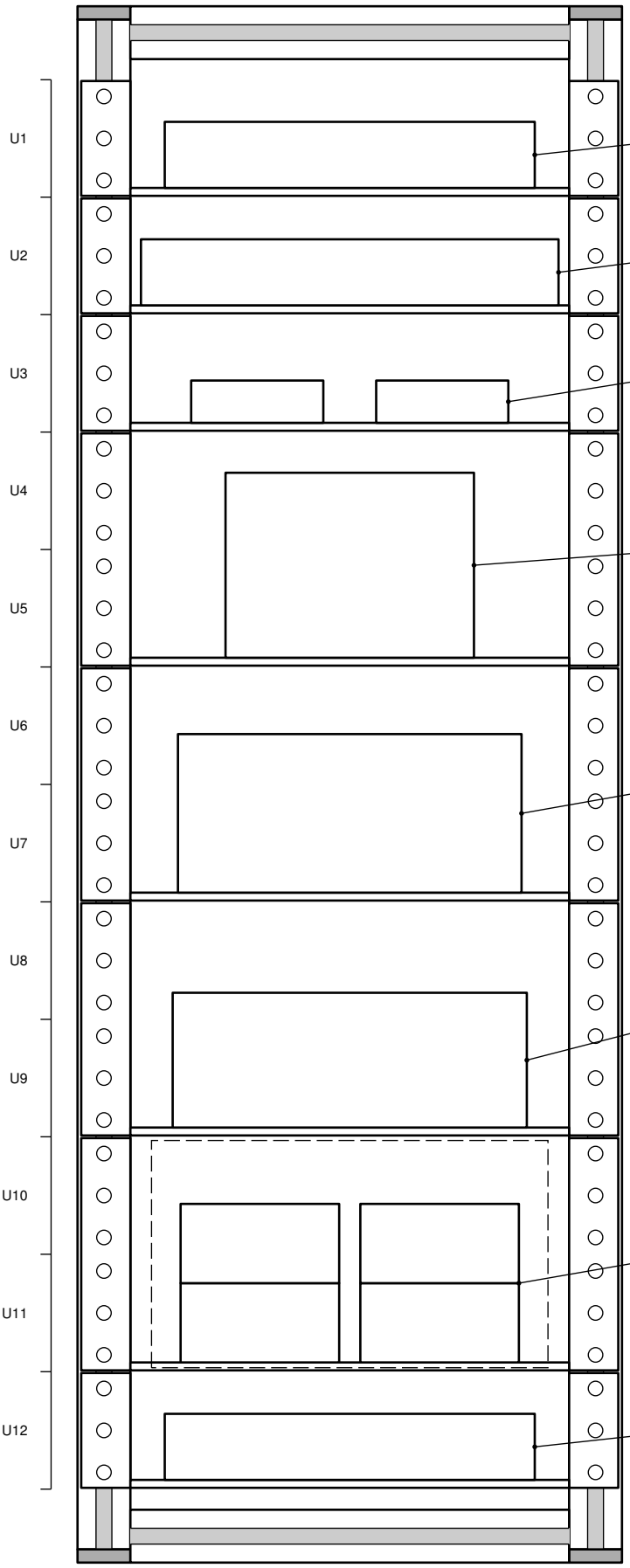
NOTES

1. A: seen from inside the frame; the cut ends of the rails show the 2020 profile section.
2. B: seen from outside the frame; the outer faces of the upright and rails are flush.
3. Corner connector drawn schematically. Item numbers as on sheets 01 to 03.

OWNER				
LUIDLAB · MINIRACK PROJECT · 8 U. · 20 × 20 ALUMINUM EXTRUSION				
TITLE **CORNER DETAILS** Top back right vertex, inside and outside views			PROJECTION **ISOMETRIC**	UNIT **mm**
MATERIAL Aluminum profile 6063-T5, 20 series (20 × 20), 6 mm slot			STANDARDS ISO 5457 · 7200 · 128 · 129-1	
DRAWING NO. **MR2020-04**	SCALE **1:1**	SHEET **4 / 5**	DATE 2026-09-13	REV. **A**

RACK UNIT LAYOUT

FRONT VIEW 1:2.5



U1 EXHAUST
2× fan, 140 × 140 × 25

U2 SWITCH
TP-Link TL-SG108, 158 × 101 × 25

U3 SBC PAIR
2× Orange Pi Zero 3, 50 × 55 × 16

U4-5 NAS
SBC + Penta SATA HAT + 4× SSD, 94 × 175 × 70

U6-7 MINI PC
AOOSTAR GEM12, 130 × 130 × 60

U8-9 GPU
Lying card via OCuLink, ≤250 × 134 × 51

U10-11 POWER
ATX PS/2 at the rear (dashed) + 4 bricks, est.

U12 INTAKE
2× fan, 140 × 140 × 25

OWNER

LUIDLAB · MINIRACK PROJECT · 8 U. · 20 × 20 ALUMINUM EXTRUSION

TITLE	PROJECTION	UNIT
RACK UNIT LAYOUT Front view, tray and equipment allocation per U		**mm**

MATERIAL	STANDARDS
Aluminum profile 6063-T5, 20 series (20 × 20), 6 mm slot	ISO 5457 · 7200 · 128 · 129-1

DRAWING NO.	SCALE	SHEET	DATE	REV.
MR2020-05	**1:2.5**	**5 / 5**	2026-09-13	**A**